CS 581 — Nuclear Cybersecurity

Security for Cyber-Physical & Nuclear Systems

Boise State University — Fall 2026

Course Information

Location: Online – Canvas; **Time**: Asynchronous
Virtual Office Hours: Wed 11am–3pm MT / By Appointment
Instructor: Christopher Spirito
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Course Objectives and Learning Outcomes

Security is paramount to creating software and systems that operate in high-consequence environments. This course introduces the fundamentals of cybersecurity and security engineering through the lens of the nuclear industry — one of the most regulated, adversarially targeted, and safety-critical sectors in the world.

Students will move beyond passive study: every module includes a hands-on **coding assignment** in which students use an **AI Coding Assistant** (e.g., Claude Code, Codex, Antigravity, or equivalent) to design, build, and iterate on working **security prototypes**. These prototypes — threat models, access control implementations, anomaly detection sketches, compliance analysis tools, and incident response artifacts — form a professional portfolio that students carry out of the course.

Topics include vulnerability analysis, common exploits and defenses, applied cryptography, ICS/SCADA protocol security, nuclear command and control, insider threat, side-channel attacks, and emerging challenges at the intersection of AI and nuclear systems. The course consists of pre-recorded lectures, AI Coding Assistant workshops, individual and group coding activities, a mini-research experience, a semester project, and practitioner-led Zoom sessions.

Course Topics

The course follows a deliberate eight-phase progression across 16 weeks, building from security foundations through to advanced and emerging threats.

Phase 1 — Foundations *(Wks 1–2)*

- Computer Security Overview
- Psychology of Security

Phase 2 — Threat Landscape *(Wks 3–4)*

- Opponents & Adversary Taxonomy
- Insider Threat

Phase 3 — Technical Core *(Wks 5–7)*

- Cryptography & Protocol Security
- Access Control & Identity
- ICS/SCADA & Digital I&C

Phase 4 — Nuclear Context *(Wks 8–9)*

- Physical Protection & Regulatory Frameworks
- Nuclear Command & Control
- Economics of Security

Phase 5 — Defense & Detection *(Wks 11–12)*

- Monitoring, Metering & Anomaly Detection
- Supply Chain Security

Phase 6 — Incident Response *(Wk 13)*

- IR in Nuclear Environments
- Zimmerman's 10 Rules for IR

Phase 7 — Advanced & Emerging *(Wks 14–15)*

- Side Channels
- AI, Autonomy & SMRs
- Emerging Threats

Phase 8 — Capstone *(Wk 16)*

- Ethics & Policy
- Guest Practitioner Perspectives

Texts and Materials

Required
Security Engineering — Ross Anderson (2nd and 3rd ed.)
Access options (no purchase required):

- **BSU Library (full text):** research.ebsco.com — log in with your BSU credentials
- **Author's website (selected chapters):** cl.cam.ac.uk/~rja14/book.html

Supplemental
Nuclear-sector primary sources assigned per module, including NRC regulatory guidance (10 CFR 73.54, RG 5.71), NEI 08-09, IAEA Nuclear Security Series documents, NIST SP 800-82, and selected academic papers on ICS/SCADA and cyber-physical system security.

Required Software
All coding workshops require an AI Coding Assistant with function calling and web search capability. Obtain access to one before Week 1:

- **Claude Code** (Anthropic) — claude.ai/code
- **Codex** (OpenAI) — platform.openai.com
- **Antigravity** (Google) — antigravity.google

All three are US-based providers with documented data handling policies appropriate for coursework. A GitHub account is also required; see Module 1 for repository setup instructions.

Course Requirements

All assessments are real work products.

Participation
Attend and engage in at least 8 live Zoom sessions with invited expert practitioners.

Ethics Paper
Personalized ethical dilemma issued by the instructor in Week 14, drawn from each student's W1–W8 portfolio artifacts. Students argue a defensible position using primary sources within a 10-day window.

Workshops & Coding Assignments
AI Coding Assistant-driven security prototypes submitted with session logs each module. Students select a target system from an eleven-system nuclear plant architecture in Week 1 and apply each subsequent workshop to a chosen system; a sequencing plan (graded separately) ensures all eleven systems appear in the portfolio by Week 14.

Portfolio Capstone
Integrated executive-level security assessment synthesizing all nine workshop artifacts into a coherent argument about nuclear plant security posture. Both repos must be in professional final state by the submission deadline.

Practitioner Engagement
Written analysis connecting each guest speaker's experience to course material.

Midterm Scenario Exercise
Personalized nuclear incident scenario at Week 10, tailored to each student's W1–W5 portfolio. Students produce a full incident response package within 72 hours.

Component	**Points**
System Selection & Sequencing Plan	30
Workshops W1–W9 (9 × 30 pts)	270
Midterm Scenario Exercise	100
Ethics Paper	150
Portfolio Capstone	250
Participation	100
Practitioner Engagement	100
Total	**1000**

Grade Policies

Submission

All assignments must be committed to the student's two course GitHub repositories (primary: technical artifacts; secondary: role analyses) by 11:59pm MT on the due date. We evaluate the last commit before the deadline — no Canvas file uploads are required for workshops. Setup instructions are in Module 1. Late submissions are subject to the following progressive penalty:

Lateness	Penalty
Up to 24 hours	10% deduction
24–48 hours	25% deduction
48–72 hours	50% deduction
72+ hours	Not accepted

Points	Grade
>899	A
800–899	B
700–799	C
600–699	D
<600	F

Grades are computed as a direct unweighted sum of all course components. Numerical grades are rounded to the nearest whole number (799.5 becomes 800, a B; 799.4 remains 799, also a B). Curves, if applied, will only ever move in your favor. Grade boundaries may be relaxed but never tightened. Grade summaries are posted on Canvas — review your scores and report any discrepancies promptly. Individual assignments may contain optional extra credit components; no separate extra credit assignments will be offered.

Warning

This course covers offensive security techniques with the objective of understanding how to design defenses. **Do not apply any technique, tool, or knowledge from this course against any real system without explicit prior written permission.** Unauthorized access is illegal, violates BSU network policy, and is incompatible with the professional standards of this field. When in doubt, stop and ask.

Course Conduct

Treat all members of this course — peers and instructor — with respect. Engage with course materials consistently, meet deadlines, and communicate professionally in all course-related interactions. For email or messages, include a clear subject line and state your purpose concisely. Active participation in Zoom sessions with guest practitioners is expected and contributes directly to your grade. Use professional language in all forums.

Syllabus Changes

This syllabus is a guide, not a contract. Deadlines, requirements, and course structure are subject to change at the instructor's discretion. Changes will be announced via Canvas and BSU email — check both regularly.